33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F16-SR-002
415V FLOTATION SWITCHROOM

F16-TR-004
TRANSFORMER ROOM

F16-TR-004
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

F16-SW-004
FLOTATION 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
2.5VA
5P20

0F-01
4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING CHARGE SUPPLY
AFD
ACB
4000A
50
51
50G
51G
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
4000 /5A
30VA
C11.0
0F-08
125A
160A
SD
NOTE 6

415V 3NE~ 50Hz 4000A 65kA for 1sec

1R-01 1R-09 1R-17 5F-01 5F-09 5F-17 INC 3F-17 3R-01 6F-01 6F-15

20A 20A 20A 20A 20A 20A 400A 400A 20A 32A
2A 32A
70A 70A 70A 70A 70A 70A 400A 400A 70A 70A
EOL

CONTINUED ON SHT 2
2000-F16-SLD-EL-10202

FEEDER TO F16-SW-104
FLOTATION 415V
ESSENTIAL SWITCHBOARD
2000-F16-SLD-EL-10251
2000-F16-SCM-EL-10246

AHF

E-SHUT
F16SR002-LCS
NOTE 12

M 3~
0.75kW
F16-FN-497
ROUGHER FLOTATION CELL 1 GEARBOX COOLING FAN
2000-F16-SCM-EL-10239

M 3~
0.75kW
F16-FN-498
ROUGHER FLOTATION CELL 2 GEARBOX COOLING FAN
2000-F16-SCM-EL-10240

M 3~
0.75kW
F16-FN-499
ROUGHER FLOTATION CELL 3 GEARBOX COOLING FAN
2000-F16-SCM-EL-10241

M 3~
0.75kW
F16-FN-500
SCAVENGER FLOTATION CELL 1 GEARBOX COOLING FAN
2000-F16-SCM-EL-10242

M 3~
0.75kW
F16-FN-501
SCAVENGER FLOTATION CELL 2 GEARBOX COOLING FAN
2000-F16-SCM-EL-10243

M 3~
0.75kW
F16-FN-502
SCAVENGER FLOTATION CELL 3 GEARBOX COOLING FAN
2000-F16-SCM-EL-10244

300A
F16-HF-004
ACTIVE HARMONIC FILTER
2000-F16-SCM-EL-10206

M 3~
0.75kW
F16-FN-503
SCAVENGER FLOTATION CELL 4 GEARBOX COOLING FAN
2000-F16-SCM-EL-10245

M 3~
7.5kW
F17-HU-422
FLOTATION TAILINGS THICKENER FEED VALVE HYDRAULIC POWER UNIT
2000-F17-SCM-EL-10202

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-004 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F16-SLD-EL-10201_Rev2

Rougher & scavenger flotation F16-SW-004 415V
MCC single line diagram sheet 1